Foundations: Itô Calculus and the SDE Framework
`hasklib` documentation

Hubert Kasinski

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Abstract

This note collects the stochastic-calculus results that the `stochastic/` layer of `hasklib` relies on: the probability-space setup, Brownian motion, the general Itô SDE $\mathrm{d}X_t = a(t, X_t)\mathrm{d}t + b(t, X_t)\mathrm{d}W_t$ (encoded by the `StochasticProcess` interface), Itô's lemma, and quadratic variation. Downstream files (`gbm.tex`, `ou.tex`, `abm.tex`, `cir.tex`, `schemes.tex`) cite the results established here. Standard measure-theoretic probability is assumed; see Shreve (2004) or Williams (1991) for construction.

1 Probability space and Brownian motion

We work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions. The filtration $\{\mathcal{F}_t\}$ represents the information available up to time t.

Definition 1 (Brownian motion)**.** A standard Brownian motion $\{W_t\}_{t\geq 0}$ is a process with

1. $W_0 = 0$ almost surely;
2. Independent increments: for $s < t < u$, $W_u - W_t$ is independent of $W_t - W_s$;
3. Gaussian increments: $W_t - W_s \sim \mathcal{N}(0, t-s)$;
4. Continuous sample paths.

Remark 1. In code, property (3) is what `NormalRng::draw()` supplies: a step of length Δt uses $W_{t+\Delta t} - W_t = \sqrt{\Delta t}\, Z$ with $Z \sim \mathcal{N}(0, 1)$. This substitution is the single bridge between the continuous theory here and the discrete simulation code.

Although the paths are continuous, they are extremely rough. Fix t and consider the difference quotient over a step $h > 0$. By the Gaussian-increment property, $W_{t+h} - W_t \sim \mathcal{N}(0, h)$, equivalently $\sqrt{h}\, Z$ with $Z \sim \mathcal{N}(0, 1)$, so

$$\frac{W_{t+h} - W_t}{h} \overset{d}{=} \frac{1}{\sqrt{h}}\, Z,$$

whose standard deviation $1/\sqrt{h}$ diverges as $h \downarrow 0$. Informally, the path moves by order $\sqrt{h}$ over a time step of order h, so its instantaneous slope is of order $1/\sqrt{h} \to \infty$ and the quotient converges to no finite limit. This heuristic is made rigorous by the Paley–Wiener–Zygmund theorem (Paley, Wiener & Zygmund 1933): with probability one, $t \mapsto W_t$ is nowhere differentiable.

Equivalently, the path has infinite first-order (total) variation on every interval, so integrals $\int f\, \mathrm{d}W_s$ cannot be defined pathwise as Riemann–Stieltjes integrals. Its *quadratic* variation, however, is finite — the content of section 2 — and it is precisely this finite second-order variation that the Itô integral and Itô's lemma are built on.

2 Quadratic variation

The property that distinguishes stochastic from ordinary calculus is that Brownian motion accumulates *quadratic* variation at a nonzero rate.

Proposition 1 (Quadratic variation of Brownian motion). *For a standard Brownian motion, the quadratic variation over* $[0,T]$ *satisfies*

$$\langle W\rangle_T \;=\; \lim_{\|\Pi\|\to 0}\sum_k \left(W_{t_{k+1}} - W_{t_k}\right)^2 \;=\; T \qquad \text{in } L^2,$$

where Π *is a partition of* $[0,T]$. *Heuristically this is written*

$$(\mathrm{d}W_t)^2 = \mathrm{d}t.$$

Proof. Write the partition as $\Pi = \{0 = t_0 < t_1 < \cdots < t_n = T\}$ and abbreviate

$$\Delta t_k = t_{k+1} - t_k, \qquad \Delta W_k = W_{t_{k+1}} - W_{t_k}, \qquad S_\Pi = \sum_{k=0}^{n-1} \left(\Delta W_k\right)^2.$$

"Convergence in L^2" means the mean-square error vanishes: $\mathbb{E}\big[(S_\Pi - T)^2\big] \to 0$ as the mesh $\|\Pi\| := \max_k \Delta t_k \to 0$. We use only two facts about the increments, both directly from the definition of Brownian motion — each $\Delta W_k \sim \mathcal{N}(0, \Delta t_k)$, and increments over disjoint intervals are independent. Nothing measure-theoretic enters beyond these.

Step 1 — the mean is exactly T. Since $\mathbb{E}[\Delta W_k] = 0$, the second moment is the variance:

$$\mathbb{E}\big[(\Delta W_k)^2\big] = \operatorname{Var}(\Delta W_k) = \Delta t_k.$$

Summing over k,

$$\mathbb{E}[S_\Pi] = \sum_k \mathbb{E}\big[(\Delta W_k)^2\big] = \sum_k \Delta t_k = T,$$

which holds for *every* partition, not merely in the limit. Because the mean is exactly T, the mean-square error *is* the variance:

$$\mathbb{E}\big[(S_\Pi - T)^2\big] = \operatorname{Var}(S_\Pi).$$

So the whole statement reduces to showing $\operatorname{Var}(S_\Pi) \to 0$.

Step 2 — variance of a single term. Normalise the increment as $\Delta W_k = \sqrt{\Delta t_k}\, Z$ with $Z \sim \mathcal{N}(0,1)$, so $(\Delta W_k)^2 = \Delta t_k\, Z^2$ and

$$\operatorname{Var}\left((\Delta W_k)^2\right) = (\Delta t_k)^2\, \operatorname{Var}(Z^2).$$

For a standard normal, $\mathbb{E}[Z^2] = 1$ and $\mathbb{E}[Z^4] = 3$, hence $\operatorname{Var}(Z^2) = \mathbb{E}[Z^4] - \left(\mathbb{E}[Z^2]\right)^2 = 3 - 1 = 2$. (The fourth moment is one Gaussian integration by parts: with the density ϕ satisfying $\phi'(z) = -z\,\phi(z)$, integrating $\int z^4 \phi\, \mathrm{d}z$ by parts gives $\mathbb{E}[Z^4] = 3\,\mathbb{E}[Z^2] = 3$.) Therefore

$$\operatorname{Var}\left((\Delta W_k)^2\right) = 2\,(\Delta t_k)^2.$$

Step 3 — variance of the sum, then let the mesh shrink. The increments ΔW_k are independent, so the squared increments $(\Delta W_k)^2$ are independent too, and the variance of a sum of independent variables is the sum of the variances:

$$\operatorname{Var}(S_\Pi) = \sum_k \operatorname{Var}\left((\Delta W_k)^2\right) = 2\sum_k (\Delta t_k)^2.$$

Bound one factor of Δt_k in each term by the mesh:

$$\sum_k (\Delta t_k)^2 \le \Big(\max_k \Delta t_k\Big) \sum_k \Delta t_k = \|\Pi\|\, T.$$

Combining,

$$\mathbb{E}\big[(S_\Pi - T)^2\big] = \operatorname{Var}(S_\Pi) \le 2\,\|\Pi\|\, T \xrightarrow[\|\Pi\|\to 0]{} 0,$$

which is exactly the claimed L^2 convergence. □

Remark 2. The mean computation ($\mathbb{E}[S_\Pi] = T$) is what makes $(\mathrm{d}W_t)^2 = \mathrm{d}t$ *plausible*; the variance computation is what makes it *usable*. Per step the squared increment $(\Delta W_k)^2$ still fluctuates — its standard deviation $\sqrt{2}\,\Delta t_k$ is the same order as its mean — but those fluctuations are independent across steps, so summing averages them out at rate $\|\Pi\|$. Inside a sum (equivalently, an integral) the random $(\Delta W_k)^2$ may therefore be replaced by the deterministic Δt_k. That replacement is the entire content of the shorthand $(\mathrm{d}W_t)^2 = \mathrm{d}t$ used in the next section.

3 The Itô stochastic differential equation

Definition 2 (Itô SDE)**.** An Itô process X_t solves the stochastic differential equation

$$\mathrm{d}X_t \;=\; a(t, X_t)\,\mathrm{d}t \;+\; b(t, X_t)\,\mathrm{d}W_t, \tag{1}$$

interpreted as the integral equation

$$X_t = X_0 + \int_0^t a(s, X_s)\,\mathrm{d}s + \int_0^t b(s, X_s)\,\mathrm{d}W_s,$$

where the second integral is an Itô integral. We call a the *drift* and b the *diffusion* coefficient.

Remark 3 (Correspondence with the code). Equation (1) is exactly what the `StochasticProcess` base class encodes: `drift(t,x)` returns $a(t, x)$ and `diffusion(t,x)` returns $b(t, x)$. Every concrete process (GBM, OU, CIR) is a choice of these two functions.

Assumption 1 (Existence and uniqueness)**.** We assume a and b satisfy the standard Lipschitz and linear-growth conditions guaranteeing a unique strong solution to (1); see Øksendal (2003), Thm. 5.2.1. *(Cited, not proved — the library never depends on the proof, only on the conclusion.)*

4 Itô's lemma

Itô's lemma is the chain rule for functions of an Itô process, and it is the engine behind every closed-form solution used in `hasklib` (the GBM exact sampler, the OU solution).

Theorem 1 (Itô's lemma, one dimension)**.** *Let X_t solve* (1) *and let $f(t, x) \in C^{1,2}$. Then*

$$\mathrm{d}f(t, X_t) = \Big(f_t + a\, f_x + \tfrac{1}{2} b^2 f_{xx}\Big)\mathrm{d}t \;+\; b\, f_x\,\mathrm{d}W_t, \tag{2}$$

where subscripts denote partial derivatives evaluated at (t, X_t).

Heuristic derivation. This is the standard second-order Taylor argument. It is not a proof in the strict sense — a rigorous version requires the construction of the Itô integral (Shreve 2004, Ch. 4), which is outside the scope of this note — but it exposes exactly where the extra term comes from, and it is all the library needs.

The one idea is this: because $\mathrm{d}X$ carries a Brownian term of size $\sqrt{\mathrm{d}t}$, a contribution that ordinary calculus discards as "second order" is in fact first order and must be kept.

Orders of magnitude. Over a step of length $\mathrm{d}t$ the increment $\mathrm{d}W_t$ has typical size $\sqrt{\mathrm{d}t}$ (it is $\mathcal{N}(0, \mathrm{d}t)$). Ranking the elementary products by their power of $\mathrm{d}t$:

$$\mathrm{d}t = O(\mathrm{d}t), \qquad \mathrm{d}W_t = O(\mathrm{d}t^{1/2}), \qquad (\mathrm{d}W_t)^2 = O(\mathrm{d}t), \qquad \mathrm{d}t\,\mathrm{d}W_t = O(\mathrm{d}t^{3/2}), \qquad (\mathrm{d}t)^2 = O(\mathrm{d}t^2).$$

Discard everything of order smaller than $\mathrm{d}t$, and use proposition 1 to replace $(\mathrm{d}W_t)^2$ by its in-sum limit $\mathrm{d}t$. This gives the *Itô multiplication table*

$$(\mathrm{d}W_t)^2 = \mathrm{d}t, \qquad \mathrm{d}t\,\mathrm{d}W_t = 0, \qquad (\mathrm{d}t)^2 = 0. \tag{3}$$

The last two entries are not literally zero; they are negligible next to $\mathrm{d}t$ and vanish in the mesh limit for exactly the reason the fluctuations did in section 2.

Taylor expansion. Expand $f(t + \mathrm{d}t,\, X_t + \mathrm{d}X)$ about (t, X_t) to second order:

$$\mathrm{d}f = f_t\,\mathrm{d}t + f_x\,\mathrm{d}X + \tfrac{1}{2} f_{tt}\,(\mathrm{d}t)^2 + f_{tx}\,\mathrm{d}t\,\mathrm{d}X + \tfrac{1}{2} f_{xx}\,(\mathrm{d}X)^2 + \cdots$$

Substitute the SDE $\mathrm{d}X = a\,\mathrm{d}t + b\,\mathrm{d}W_t$ and square:

$$(\mathrm{d}X)^2 = a^2(\mathrm{d}t)^2 + 2\,ab\,\mathrm{d}t\,\mathrm{d}W_t + b^2(\mathrm{d}W_t)^2.$$

Apply (3). In $(\mathrm{d}X)^2$ the first two terms die and only $b^2(\mathrm{d}W_t)^2 = b^2\,\mathrm{d}t$ survives:

$$(\mathrm{d}X)^2 = b^2\,\mathrm{d}t.$$

The two remaining second-order terms of the expansion vanish for the same reason:

$$\tfrac{1}{2} f_{tt}\,(\mathrm{d}t)^2 = 0, \qquad f_{tx}\,\mathrm{d}t\,\mathrm{d}X = f_{tx}\big(a\,(\mathrm{d}t)^2 + b\,\mathrm{d}t\,\mathrm{d}W_t\big) = 0.$$

What remains is

$$\mathrm{d}f = f_t\,\mathrm{d}t + f_x\,\mathrm{d}X + \tfrac{1}{2} f_{xx}\,b^2\,\mathrm{d}t.$$

Substitute $\mathrm{d}X = a\,\mathrm{d}t + b\,\mathrm{d}W_t$ once more and collect the $\mathrm{d}t$ and $\mathrm{d}W_t$ parts:

$$\mathrm{d}f = \Big(f_t + a\,f_x + \tfrac{1}{2} b^2 f_{xx}\Big)\mathrm{d}t + b\,f_x\,\mathrm{d}W_t,$$

which is (2). □

Remark 4. Had X been a smooth (finite-variation) path, $(\mathrm{d}X)^2$ would have been genuinely $O((\mathrm{d}t)^2)$ and dropped out, leaving the ordinary chain rule $\mathrm{d}f = f_t\,\mathrm{d}t + f_x\,\mathrm{d}X$. The whole gap between stochastic and ordinary calculus is the single surviving term $\frac{1}{2} b^2 f_{xx}\,\mathrm{d}t$, and it survives only because $(\mathrm{d}W_t)^2 = \mathrm{d}t$ rather than 0 — i.e. because Brownian motion has nonzero quadratic variation. Every closed-form result in `hasklib` that "corrects a drift by half a variance" (the $-\frac{1}{2}\sigma^2$ in the GBM log-solution, the Itô term in the OU solution) is exactly this one term.

Remark 5. The single downstream consequence to keep in view: applying theorem 1 with $f(t, x) = \ln x$ to geometric Brownian motion linearises the SDE and yields the exact terminal law sampled by `GBM::sample_terminal`. That derivation is carried out in `gbm.tex`.

5 What the following files derive

Each companion file starts from (1) with a specific (a, b) and uses theorem 1 to produce results the code depends on:

- `gbm.tex`: $a = \mu x$, $b = \sigma x \Rightarrow$ exact solution and moments (verified in `test_stochastic.cpp`).
- `ou.tex`: $a = \kappa(\theta - x)$, $b = \sigma \Rightarrow$ Gaussian solution, mean/variance, stationary law.
- `abm.tex`: $a = \mu$, $b = \sigma \Rightarrow$ exact solution by direct integration (no Itô's lemma needed); the $\kappa = 0$ degenerate case of `ou.tex`, plus a drift.
- `cir.tex`: $a = \kappa(\theta - x)$, $b = \sigma\sqrt{x} \Rightarrow$ moments, Feller condition, motivation for full truncation.
- `schemes.tex`: discretisation of (1) — Euler–Maruyama (strong order 0.5) and Milstein (strong order 1.0) via stochastic Taylor expansion (Kloeden & Platen 1992).

References

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